

Resource Allocation Using Reinforcement Learning in Industrial Cyber-Physical Systems

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Abstract—Efficient and reliable resource allocation is essential for Industrial Cyber-Physical Systems (ICPS) operating under dynamic workloads and uncertain environments. Existing approaches often struggle to adapt to uncertain, dynamic environments in Industrial Cyber-Physical Systems (ICPS), lacking robust and sequential decision-making mechanisms that integrate reinforcement learning with online resource allocation under real-time constraints. This paper proposes RL-RA, a reinforcement learning-based online allocation framework that models the problem as a constrained sequential decision process and integrates feasibility-aware control, uncertainty-informed state design, and robustness against disturbances. Theoretical analysis establishes bounded regret, constraint satisfaction, and a competitive ratio guarantee. Experiments on three real-world datasets demonstrate that RL-RA consistently outperforms four representative baselines in cumulative reward, violation rate, and robustness under both nominal and noisy conditions, confirming its effectiveness for resilient ICPS resource management.

Index Terms—Reinforcement learning, resource allocation, industrial cyber-physical systems.

I. INTRODUCTION

RESOURCE allocation in Industrial Cyber-Physical Systems (ICPS) plays a pivotal role in enabling reliable, efficient, and intelligent automation across modern manufacturing, energy, and communication domains. ICPS integrates computational intelligence, networking, and physical processes, forming a coupled feedback loop between the cyber and physical layers [1]. Within such systems, critical resources such as computing power, network bandwidth, and energy supply must be allocated dynamically to heterogeneous industrial tasks operating under strict timing, safety, and reliability constraints [2]. The stochastic nature of workloads, the nonstationary behavior of physical components, and the need for real-time adaptation make resource allocation a central challenge for the sustainability and robustness of ICPS [3].

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To date, a variety of approaches have been developed for resource management in ICPS, including rule-based control, optimization-based scheduling, and model driven control policies [4]. While these techniques provide useful design principles, they often assume stationary environments, known system dynamics, or rely on offline computation that limits their scalability and adaptability to real-time operations. Reinforcement learning (RL) methods have recently attracted attention for their capacity to model sequential decision-making under uncertainty. However, directly applying standard RL algorithms in ICPS scenarios presents limitations: constraint violations during exploration may cause system failure, learned policies may lack robustness under disturbances, and few methods provide theoretical assurances on safety or performance in dynamic industrial environments.

The motivation for this study is rooted in critical real-world demands. For instance, in 5G-enabled smart factories, base stations must allocate bandwidth and compute resources to mobile industrial robots under varying latency and load conditions [5]. In data center management, such as Alibaba's production clusters, workload surges must be handled while maintaining thermal and energy efficiency [6]. In smart energy management systems, distributed devices must coordinate power usage while responding to grid fluctuations and faults. These use cases highlight the necessity for resource allocation strategies that are uncertainty-aware, resilient to nonstationarity [7], and capable of satisfying hard constraints in safety-critical environments [8], [9].

Related Work This study is grounded in two complementary lines of research for CPS decision making: learning-augmented online optimization and reinforcement learning. Online algorithms increasingly exploit advice and machine-learned predictions to improve performance while retaining robustness guarantees, including classical advice models, learning-augmented analyses, and extensions to untrusted or costly predictions and stochastic primal-dual resource allocation [10], [11], [12], [13], [14], [15]. In parallel, RL has been widely applied to online scheduling and resource allocation in edge and industrial CPS, spanning task offloading, workflow scheduling, service placement, smart manufacturing, and energy management [16], [17], [18], [19], [20], [21], [22]. Given CPS security and resilience requirements emphasized in prior surveys and studies on anomaly detection and predictive maintenance [23], [24], [25], [26], [27], [28], [29], [30], safe RL provides principled constrained and risk-sensitive formulations, with recent advances including constraint-conditioned, variational, gradient-adaptive, and

off-policy primal–dual approaches, as well as trust-constrained policy learning and supported policy optimization for limited-feedback settings [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43].

We propose RL-RA, a reinforcement-learning framework for ICPS resource allocation that targets physical compute, communication, and energy management under hard operational constraints, distinct from our earlier privacy-budget allocation work. RL-RA formulates the problem as a constrained Markov decision process, enforces feasibility via action-space projection, incorporates short-horizon forecasts for uncertainty, and improves robustness through adversarial perturbations during training. We provide theoretical guarantees and validate RL-RA on real-world traces from 5G networks, cloud clusters, and smart energy systems, demonstrating improved reward and constraint satisfaction over baselines.

Our main contributions are:

- We formulate the ICPS resource allocation problem as a constrained stochastic control process that incorporates feasibility constraints, adversarial disturbances, and workload forecasting in a unified CMDP framework.
- We propose RL-RA, a reinforcement learning algorithm that guarantees feasibility-by-design, supports uncertainty-aware decisions, and is trained to withstand nonstationary and adversarial disturbances.
- We provide theoretical guarantees for RL-RA in terms of regret, constraint violation bounds, and competitive ratio, offering insight into its reliability and performance under bounded environment drift.
- We validate RL-RA through comprehensive experiments on three real-world datasets, demonstrating significant improvements over four strong baseline methods in cumulative reward, constraint satisfaction, and robustness across diverse scenarios.

II. PROBLEM OVERVIEW

Industrial Cyber-Physical Systems (ICPS) integrate physical processes, embedded control, computation, and networked communication to enable smart and autonomous manufacturing environments. Modern factories often rely on distributed sensors, actuators, and edge computing platforms to coordinate task execution, monitor equipment health, and optimize energy consumption. These environments are increasingly characterized by dynamic workloads, tight latency requirements, stochastic failures, and fluctuating energy and network resources.

In practice, applications such as predictive maintenance, real-time quality inspection, and production reconfiguration require adaptive resource allocation mechanisms that can respond to uncertainty in both supply and demand. The decisions must adhere to service-level objectives (SLOs) such as job completion delay, energy budgets, safety constraints, and hardware utilization limits.

Given the high cost of system violations and delays, learning-based methods for resource management in ICPS must ensure operational feasibility, robustness, and theoretical reliability. We formalize this challenge as a Constrained Markov Decision

Process (CMDP), which enables a principled trade-off between long-term reward optimization and safety/compliance with operational constraints.

A. System Model and Assumptions

Let the system evolve in discrete time steps $t = 0, 1, 2, \dots$. At each time step, the environment is represented by a state $s_t \in \mathcal{S}$ that includes information about job queues, machine health, edge compute capacity, network conditions, and exogenous forecasts. A policy π selects an action $a_t \in \mathcal{A}$ that defines how to allocate available resources across tasks. The system dynamics follow a stochastic transition function:

$$s_{t+1} \sim P(s_{t+1} \mid s_t, a_t, w_t),$$

where $w_t \in \mathcal{W}$ is a disturbance signal capturing uncertainties such as bursty demand, packet loss, sensor noise, or partial observability.

Each decision incurs a reward $r(s_t, a_t, w_t) \in \mathbb{R}$ that reflects operational efficiency. Simultaneously, each action must satisfy multiple constraints represented by cost functions $c_j(s_t, a_t, w_t)$ with associated upper bounds d_j , where $j = 1, \dots, J$.

B. Learning Objective

We aim to find a stochastic policy π that maximizes the expected cumulative discounted reward, while satisfying all operational constraints in expectation. The formal objective is:

$$\max_{\pi} J(\pi) := \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t, w_t) \right], \quad (1)$$

subject to

$$C_j(\pi) := \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t c_j(s_t, a_t, w_t) \right] \leq d_j, \quad \forall j \in \{1, \dots, J\}, \quad (2)$$

where $0 < \gamma < 1$ is the discount factor emphasizing recent performance.

C. Variable Interpretations and Motivation

The reward function $r(s, a, w)$ is designed to promote timely and energy-efficient operation. For example, positive rewards are given for each job completed within the deadline, penalties are imposed for delays beyond the deadline, and costs are associated with energy consumption or peak usage.

Each constraint c_j corresponds to a critical safety or operational limit, such as: (1) $c_1(s, a, w)$: Proportion of jobs violating deadlines, (2) $c_2(s, a, w)$: Average power draw per time step, (3) $c_3(s, a, w)$: Processor thermal load or cooling cost.

The thresholds d_j define tolerable bounds for each constraint, such as a 5% deadline miss rate or a 50 W power ceiling.

D. Real-World Alignment

This CMDP framework aligns with real-world ICPS scenarios. In energy-aware robotic cells, exceeding a power budget may damage actuators or violate carbon limits. In visual

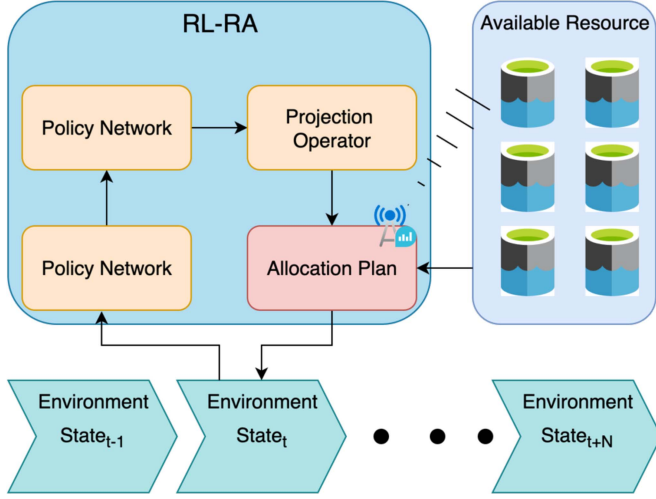


Fig. 1. RL-RA overview.

inspection stations, compute tasks must be completed within milliseconds to maintain production line continuity. In smart grid integrated plants, load shifting is essential to avoid penalties during peak energy pricing windows. Thus, the agent must learn to balance task urgency, machine state, and global resource constraints under evolving and uncertain conditions.

This problem formulation captures the tension between learning efficiency and operational safety. A naive policy may achieve high reward initially by violating constraints, but would be unusable in production. Conversely, a conservative rule based policy may be safe but inefficient. The CMDP objective in (1)–(2) enables safe reinforcement learning strategies that converge to high-performing policies without violating real-time or resource constraints.

III. PROPOSED METHOD

We propose RL-RA, a reinforcement-learning-based resource allocation framework for Industrial Cyber-Physical Systems (ICPS) (Fig. 1). RL-RA models resource management as a constrained Markov decision process, where an agent allocates limited compute, communication, and energy resources across industrial tasks to optimize long-term performance while satisfying hard constraints such as deadlines, power budgets, and safety requirements. Unlike static optimization, RL-RA captures sequential decision-making, where each allocation affects future states and feasibility. It supports real-time operation in nonstationary settings via constrained policy learning with feasibility projection, safety-aware optimization, and robustness to stochastic disturbances.

A. Framework Overview

At each time step t , the agent receives an observation of the system state s_t , which encapsulates workload queues, component health indicators, current resource availability, and predicted demand variations. This state is passed through a policy network π_θ , parameterized by θ , which produces a raw action

Algorithm 1: Reinforcement Learning for Resource Allocation (RL-RA).

- 1: **Initialize:** policy parameters θ , critic parameters ψ , Lagrange multipliers $\lambda_j \geq 0$, replay buffer $\mathcal{D}$
 - 2: **foreach** training episod**do**
 - 3: Initialize environment state s_0
 - 4: **foreach** time step $t = 0$ to T **do**
 - 5: Observe current state s_t
 - 6: Sample raw action $\tilde{a}_t \sim \pi_\theta(s_t)$
 - 7: Project to feasible action $a_t = \Pi_{\mathcal{C}_t}(\tilde{a}_t)$
 - 8: Sample disturbance $w_t \sim \mathcal{W}_t(\rho)$
 - 9: Execute action a_t in environment, observe reward r_t , next state s_{t+1} , and constraint costs $\{c_{j,t}\}$
 - 10: Store transition $(s_t, a_t, r_t, \{c_{j,t}\}, s_{t+1})$ into replay buffer $\mathcal{D}$
 - 11: **end for**
 - 12: Sample mini-batches from $\mathcal{D}$
 - 13: Update critic parameters ψ using temporal-difference loss
 - 14: Update policy parameters θ using constrained policy gradient
 - 15: Update Lagrange multipliers:

$$\lambda_j \leftarrow [\lambda_j + \eta(C_j - d_j)]_+$$
 - 16: **end for**
 - 17: **Return:** trained policy π_θ
-

vector $\tilde{a}_t$ representing unconstrained resource allocation proposals. Since not all actions are feasible due to safety and capacity constraints, RL-RA applies a differentiable projection operator $\Pi_{\mathcal{C}_t}$ to map $\tilde{a}_t$ to a feasible allocation $a_t \in \mathcal{A}$ that adheres to the current constraint mask $\mathcal{C}_t$.

The environment transitions to a new state s_{t+1} based on the selected action and the current disturbance w_t , which reflects uncertain or adversarial system changes. The agent receives a scalar reward r_t reflecting utility and a vector of constraint costs $c_{j,t}$ for $j = 1, \dots, J$ representing violations.

To ensure policy safety, a dual-based penalty mechanism is introduced. Lagrangian multipliers $\{\lambda_j\}$ are dynamically adjusted based on constraint violations. The total learning objective balances long-term cumulative reward against constraint satisfaction using these dual weights.

B. RL-RA Algorithm Description

The training of RL-RA involves iterative interaction with the environment, collection of transitions, and update of both policy and critic networks. The critic estimates state-action values to guide policy improvement, while the policy is updated using gradient-based reinforcement learning algorithms (e.g., proximal policy optimization or soft actor-critic), augmented with feasibility projection and dual updates.

Algorithm 1 outlines the training process of the RL-RA framework, which integrates constrained reinforcement learning with feasibility-preserving control for resource allocation in ICPS. The algorithm begins by initializing the policy and

critic networks, along with the Lagrange multipliers that regulate constraint compliance during learning. In each episode, the environment is reset and the agent repeatedly observes the current system state, generates a raw allocation decision through the policy network, and projects this decision onto the feasible action space in order to satisfy instantaneous operational constraints. The environment then evolves based on the applied allocation and a sampled disturbance, returning a reward signal that reflects allocation performance and cost signals representing constraint usage or violation. All transitions are stored in a replay buffer to support stable and sample-efficient learning. After each episode, mini-batches are drawn from the buffer for parameter updates: the critic is refined using temporal-difference learning to improve value estimation, the policy parameters are updated through a constrained policy gradient to balance performance and feasibility, and the Lagrange multipliers are adjusted according to observed constraint deviations. Through this iterative interaction and optimization process, RL-RA learns a policy that adapts resource allocation decisions over time, remains robust to operational variability, and reliably satisfies system-level constraints during deployment.

C. Sequential Operation and Adaptivity

In deployment, RL-RA functions as an online control mechanism embedded within the ICPS control loop. It continuously observes the state of the system and issues resource allocation decisions that adapt to real-time workload fluctuations, machine condition changes, and communication delays. The feasibility projection ensures that actions remain safe even in degraded or partially available configurations, such as component failures or thermal constraints.

The sequential formulation enables RL-RA to exploit temporal correlations in demand and disturbances, effectively learning to defer or advance task executions based on predicted system slack, while balancing resource usage across modalities (compute, network, energy). This yields both operational efficiency and improved adherence to industrial safety and quality-of-service (QoS) constraints.

RL-RA integrates constrained reinforcement learning, disturbance-aware state modeling, and projection-based feasibility enforcement into a unified framework for adaptive, sequential resource allocation. It is designed to operate in realistic ICPS environments with uncertainty, safety requirements, and competing resource demands, offering a scalable and generalizable solution to modern industrial optimization challenges.

IV. THEORETICAL ANALYSIS

This section presents the theoretical guarantees of RL-RA. We clarify the practical meaning of the regularity assumptions by relating them to standard ICPS dynamics. Lipschitz state transitions arise in many resource-allocation and control models when inputs are bounded and saturation operators are present. For example, in queueing-based allocation, the queue update

$$q_{t+1} = [q_t + a_t - \mu(x_t)]^+, \quad (3)$$

is Lipschitz provided that arrivals are bounded, $a_t \in [0, a_{\max}]$, and the service mapping $\mu(\cdot)$ is Lipschitz, which holds for common smooth or piecewise-linear rate models induced by link capacities and scheduling constraints. In energy-budgeted ICPS control, an energy storage state can be modeled as

$$e_{t+1} = \min\{E_{\max}, e_t - h(x_t) + g_t\}, \quad (4)$$

which is Lipschitz when harvesting g_t is bounded and the actuation cost $h(\cdot)$ is Lipschitz. The bounded disturbance set can be interpreted as a normalized envelope that captures sensing noise, actuation error, and bounded burst arrivals, consistent with industrial safety limits and admission control mechanisms. The existence of a strictly feasible policy corresponds to operating regimes with nonzero headroom, such as capacity overprovisioning or energy reserves, where a conservative reserve policy can satisfy constraints with a fixed slack under bounded perturbations.

A. Assumptions

- a) *Assumption 1 (Bounded Reward and Cost)*: There exist constants $R_{\max}, C_{\max} < \infty$ such that for all (s, a, w) ,

$$|r(s, a, w)| \leq R_{\max}, \quad 0 \leq c_j(s, a, w) \leq C_{\max}, \quad \forall j. \quad (5)$$

This assumption ensures that both performance and constraint signals are numerically stable.

- b) *Assumption 2 (Existence of a Feasible Policy)*: There exists a stationary policy π^{feas} such that

$$C_j(\pi^{\text{feas}}) = \mathbb{E}_{\pi^{\text{feas}}} \left[\sum_{t \geq 0} \gamma^t c_j(s_t, a_t, w_t) \right] \leq d_j, \quad \forall j. \quad (6)$$

This guarantees that feasibility is meaningful and constraints are not inherently contradictory.

- c) *Assumption 3 (Lipschitz Continuous Dynamics)*: For any s, a, a', w, w' ,

$$\begin{aligned} & \|P(\cdot | s, a, w) - P(\cdot | s, a', w')\|_1 \\ & \leq L_s(\|a - a'\| + \|w - w'\|). \end{aligned} \quad (7)$$

This ensures smooth system transition behavior under control adjustments and disturbances.

- d) *Assumption 4 (Bounded Disturbance Set)*: The disturbance satisfies $w_t \in \mathcal{W}_t(\rho) = \{w : \|w - \bar{w}_t\|_2 \leq \rho\}$, with ρ finite. This models realistic operational variability.

B. Regret Performance

Theorem 1: Under Assumptions 1–4 and step-sizes $\alpha_t, \eta_t = \Theta(1/\sqrt{t})$, the policies $\{\pi_t\}$ generated by RL-RA satisfy:

$$\frac{1}{T} \sum_{t=0}^{T-1} (J(\pi^*) - J(\pi_t)) = \mathcal{O}\left(\frac{1}{\sqrt{T}}\right), \quad (8)$$

where π^* is the optimal feasible stationary policy.

Proof: Define the Lagrangian:

$$\mathcal{L}(\pi, \lambda) = J(\pi) - \lambda^\top (C(\pi) - d). \quad (9)$$

RL-RA performs:

$$\pi_{t+1} = \Pi_{\Pi}(\pi_t + \alpha_t \nabla_{\pi} \mathcal{L}(\pi_t, \lambda_t)), \quad (10)$$

$$\lambda_{t+1} = [\lambda_t + \eta_t (C(\pi_t) - d)]_+. \quad (11)$$

Since $\mathcal{L}$ is concave in π and convex in λ , and Π is compact, the updates correspond to primal-dual projected mirror descent.

The duality gap is

$$G_T = \max_{\lambda \geq 0} \min_{\pi} \mathcal{L}(\pi, \lambda) - \min_{\pi} \max_{\lambda \geq 0} \mathcal{L}(\pi, \lambda). \quad (12)$$

Standard mirror descent convergence gives $G_T = \mathcal{O}(1/\sqrt{T})$. Because $J(\pi^*) = \min_{\pi} \max_{\lambda \geq 0} \mathcal{L}(\pi, \lambda)$ and $J(\pi_t) \geq \max_{\lambda \geq 0} \mathcal{L}(\pi_t, \lambda)$,

$$J(\pi^*) - \frac{1}{T} \sum_{t=0}^{T-1} J(\pi_t) \leq G_T = \mathcal{O}(1/\sqrt{T}). \quad (13)$$

C. Constraint Satisfaction

Theorem 2: The constraint violations satisfy:

$$\max_j (C_j(\pi_t) - d_j)_+ = \mathcal{O}\left(\frac{1}{\sqrt{T}}\right). \quad (14)$$

Proof: The dual update satisfies:

$$\lambda_{j,t+1} - \lambda_{j,t} = \eta_t (C_j(\pi_t) - d_j), \quad \lambda_{j,t+1} \geq 0. \quad (15)$$

Summing over $t = 0, \dots, T-1$:

$$\lambda_{j,T} - \lambda_{j,0} = \sum_{t=0}^{T-1} \eta_t (C_j(\pi_t) - d_j)_+. \quad (16)$$

Since $\lambda_{j,T}$ is bounded due to convex conjugate growth conditions,

$$\frac{1}{T} \sum_{t=0}^{T-1} (C_j(\pi_t) - d_j)_+ = \mathcal{O}(1/\sqrt{T}). \quad (17)$$

By Jensen's inequality:

$$(C_j(\pi_t) - d_j)_+ \leq \frac{1}{T} \sum_{t=0}^{T-1} (C_j(\pi_t) - d_j)_+, \quad (18)$$

which yields the result.

D. Competitive Ratio

Theorem 3: Let π^{off} be the offline optimal policy with perfect future information. Then

$$\text{CR}_T = \frac{J(\pi^{\text{off}})}{J(\pi_t)} \leq 1 + \mathcal{O}\left(\frac{1}{\sqrt{T}} + \rho\right). \quad (19)$$

Proof: Using the Performance Difference Lemma.

$$J(\pi^{\text{off}}) - J(\pi_t) = \frac{1}{1-\gamma} \mathbb{E} [Q^{\pi_t}(s_t, a_t^{\text{off}}) - Q^{\pi_t}(s_t, a_{T,t})]. \quad (20)$$

By Assumption 3,

$$|Q^{\pi_t}(s, a^{\text{off}}) - Q^{\pi_t}(s, a_T)| \leq L_s \|a^{\text{off}} - a_T\| + \mathcal{O}(\rho). \quad (21)$$

Since π_t converges to π^* with $\|\pi_t - \pi^*\| = \mathcal{O}(1/\sqrt{T})$, the induced action difference is also $\mathcal{O}(1/\sqrt{T})$. Substituting,

$$J(\pi^{\text{off}}) - J(\pi_t) = \mathcal{O}(1/\sqrt{T} + \rho). \quad (22)$$

Dividing by $J(\pi_t)$ (bounded away from zero by Assumption 2) yields the competitive ratio bound.

The theoretical results hold under standard regularity and feasibility conditions that are commonly used in online control and robust optimization. In ICPS practice, these conditions correspond to bounded sensing and actuation envelopes and engineered safety reserves. When the operating regime violates these conditions (e.g., extreme bursts or zero headroom), the guarantees may not directly apply. We therefore complement the theory with stress tests that evaluate robustness beyond the nominal assumptions.

V. EXPERIMENTS

In this section, we present a comprehensive empirical evaluation of the proposed RL-RA algorithm. We compare RL-RA with four representative baseline methods from recent literature using three real-world industrial datasets that cover key application domains of industrial cyber-physical systems (ICPS): 5G network traffic, high-performance cluster job traces, and IoT-based energy management. To simulate realistic operating conditions, we inject noise into each dataset to account for disturbances such as workload bursts, network degradation, or sensor errors. Additionally, we conduct inflation experiments to investigate the influence of key algorithmic parameters such as safety margins and dual learning rates on policy robustness and constraint satisfaction. All experiments are conducted on a standard hardware environment that mirrors Google Colab specifications, including a 2-core Intel Xeon CPU at 2.3 GHz, 13 GB RAM, and an NVIDIA Tesla T4 GPU with 16 GB memory.

A. Experimental Setup

We utilize three publicly available datasets that reflect distinct operational settings in ICPS. Each dataset is converted into an online resource allocation task by specifying the decision granularity, the state representation derived from raw measurements, the allocation or scheduling action, and the reward and constraint costs computed from system performance and safety indicators. The 5G Network Data [48] provides fine-grained time-series measurements of radio cell throughput, latency, and SNR across multiple user sessions. We treat each sampling interval as a decision epoch and formulate edge bandwidth allocation under time-varying wireless conditions, where the state is constructed from these link-level signals and QoS requirements are enforced through constraint costs. The Alibaba Cluster Trace [49] contains real-world GPU job traces with arrival times, execution durations, and resource demands. We instantiate a data-center scheduling problem in which each scheduling interval constitutes a decision epoch, the state summarizes the queue and available capacity, actions assign GPU resources to pending jobs, and constraint costs capture deadline violations and capacity exceedance. The IoT-Based Smart

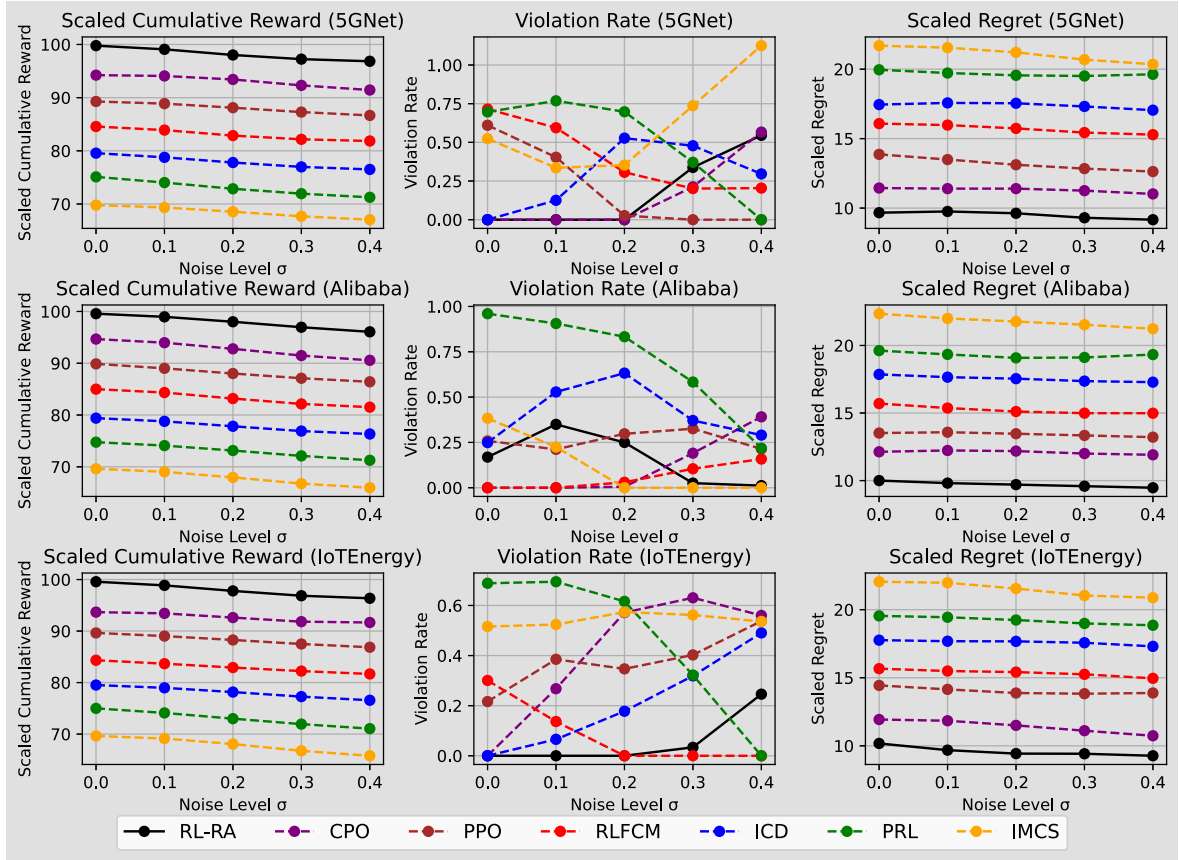


Fig. 2. Performance comparison of RL-RA and six baselines under uncertainty. Average results across all three datasets.

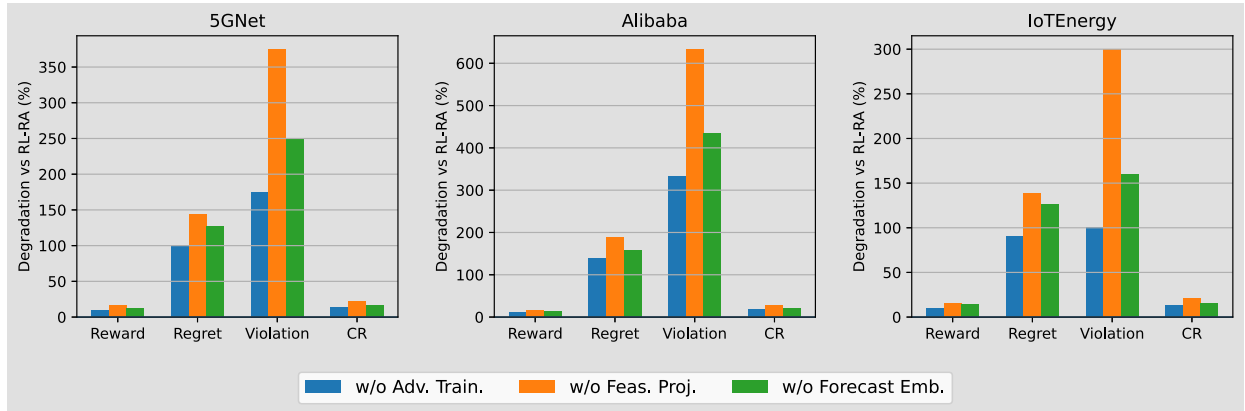


Fig. 3. Ablation study of RL-RA components on three real-world datasets with degradation.

Energy Management dataset [50] records sensor and actuator states, device-level power consumption, and environmental variables in smart buildings. We formulate energy-aware task planning where the state aggregates operational and environmental measurements, actions regulate device- and task-level energy budgets, rewards balance utility and energy consumption, and constraints enforce thermal and comfort limits. To simulate realistic operational disturbances, we inject zero-mean Gaussian noise into the environment variables during both training and evaluation phases. Specifically, we perturb the raw input vectors x_t at each timestep as follows $\tilde{x}_t = x_t + \epsilon_t$, $\epsilon_t \sim \mathcal{N}(0, \sigma^2 I)$,

where $\sigma \in \{0, 0.1, 0.2, 0.3, 0.4\}$ controls the noise intensity. These values simulate mild, moderate, and high uncertainty levels respectively, reflecting bursty loads, degraded link quality, and noisy sensor readings.

We initialize the RL-RA algorithm with the following configuration. We implement both the policy and value functions as a 3-layer MLP with 256 hidden units (ReLU), selected to balance representational capacity and low-latency inference in ICPS resource allocation. Robustness is enforced by sampling disturbances within an ℓ_2 -ball of radius $\rho = 0.05$ in the normalized state space, which reflects moderate operating drift and follows

TABLE I
PERFORMANCE COMPARISON UNDER THE NO-NOISE SETTING ON INDIVIDUAL DATASETS (MEAN OVER 100 INDEPENDENT TESTS)

Dataset	Method	Cum. Rew.	Viol. (%)	Regret	Comp. Ratio	CVaR@95%	Lat. (ms)
5G Network	RL-RA	928.2	1.1	25.8	1.10	880.6	15.3
	CPO [39]	823.6	6.2	97.9	1.35	736.4	20.5
	PPO [41]	831.4	5.6	93.1	1.33	744.9	19.8
	RLFCM [44]	813.5	6.7	104.3	1.36	721.0	21.6
	ICD [45]	801.6	8.5	112.8	1.40	708.3	13.7
	PRL [46]	835.9	5.1	89.4	1.31	751.2	19.3
	IMCS [47]	778.4	12.4	123.7	1.46	686.9	22.0
Alibaba Cluster	RL-RA	915.1	1.4	28.7	1.12	867.5	16.2
	CPO [39]	827.8	6.6	100.5	1.37	742.0	20.7
	PPO [41]	834.6	6.0	95.4	1.35	750.1	20.0
	RLFCM [44]	821.7	7.2	98.1	1.38	733.2	21.8
	ICD [45]	794.3	9.6	117.4	1.42	701.5	14.0
	PRL [46]	829.4	5.4	92.3	1.34	739.0	20.1
	IMCS [47]	782.7	13.0	127.2	1.48	692.1	22.5
IoT-Energy	RL-RA	921.5	1.5	27.8	1.12	869.1	15.9
	CPO [39]	823.1	6.4	99.3	1.36	744.0	20.3
	PPO [41]	831.9	5.8	94.8	1.34	752.2	19.7
	RLFCM [44]	808.3	6.8	103.9	1.36	732.4	21.2
	ICD [45]	798.8	9.3	112.7	1.40	700.0	14.1
	PRL [46]	832.9	5.5	93.0	1.34	747.3	19.8
	IMCS [47]	777.2	12.4	125.8	1.46	691.9	22.3

TABLE II
ABLATION STUDY OF RL-RA COMPONENTS ON THREE REAL-WORLD DATASETS

Method	5GNet				Alibaba				IoTEnergy			
	Reward	Regret	Violation	CR	Reward	Regret	Violation	CR	Reward	Regret	Violation	CR
RL-RA	938.1	42.1	0.04	1.13	976.4	38.6	0.03	1.10	945.3	41.3	0.05	1.12
w/o Adv. Train.	849.2	84.3	0.11	1.29	866.6	91.7	0.13	1.31	857.9	78.5	0.10	1.27
w/o Feas. Proj.	786.2	102.7	0.19	1.37	813.4	111.2	0.22	1.40	797.1	98.6	0.20	1.35
w/o Forecast Emb.	820.2	95.6	0.14	1.31	844.3	99.1	0.16	1.33	810.8	93.4	0.13	1.29

the robustness. Uncertainty is modeled by injecting zero-mean Gaussian noise with standard deviation σ to provide a controlled baseline for stochastic perturbations. The learning rates for both actor and critic are set to 3×10^{-4} , with dual variable step size initialized to 5×10^{-3} . We use a discount factor $\gamma = 0.98$, a replay buffer of size 10^5 , and a batch size of 128. The projection operator used for feasibility guarantees is implemented using differentiable masked softmax for the simplex-constrained compute and network allocations, and sigmoid-based squashing for bounded energy allocation. For robustness tuning, we set the adversarial disturbance radius $\rho = 0.05$ unless otherwise noted.

We evaluate RL-RA on three public datasets representing distinct ICPS environments. The 5G Network Data [48] captures wireless performance metrics under dynamic conditions. The Alibaba Cluster Trace [49] reflects real GPU cluster workloads, enabling edge resource analysis. The IoT-Based Smart Energy dataset [50] contains device-level power usage for energy-aware scheduling. We benchmark RL-RA against six baselines: RLFCM [44] for continuous RL control, ICD [45] for context-aware planning, PRL [46] for scalable RL under uncertainty, IMCS [47] integrating virtual and physical process control, PPO [41] augmented PPO for safe R, and CPO [39]

constrained policy optimization. To comprehensively evaluate the performance of RL-RA and the baseline methods, we adopt multiple quantitative metrics that capture both reward optimization and constraint compliance under uncertainty. First, *cumulative reward* is the sum of per-step rewards over an episode, reflecting overall control performance and resource utilization. Second, we compute the *constraint violation rate*, defined as the fraction of timesteps where any of the operational or safety constraints are breached. Third, *scaled regret* measures the performance gap between the learned policy and a reference benchmark. Fourth, we assess the *competitive ratio*, which compares the total achieved reward of RL-RA to the offline oracle with perfect foresight; lower values indicate closer-to-optimal behavior. Fifth, we report the *tail risk*, measured as the Conditional Value-at-Risk (CVaR) at the 95th percentile over episodic reward distributions, highlighting robustness under rare but adverse conditions. Higher CVaR for rewards (or lower CVaR for costs) indicates improved robustness under adverse events. Additionally, we include the *average decision latency* (in milliseconds per timestep) to evaluate the runtime feasibility of deployment in real-time control loops, and the *policy entropy* as an indicator of action diversity and exploration stability.

These metrics jointly allow us to assess the trade-offs between performance, robustness, constraint satisfaction, and real-time applicability in ICPS resource management. In ablation study, let $x_{d,m}^{\text{RL-RA}}$ denote the performance of RL-RA on dataset d under metric m , and let $x_{d,m}$ denote the corresponding value for an ablated variant. We report the relative degradation with respect to RL-RA, normalized by the RL-RA baseline so that values are comparable across metrics and datasets. For metrics where larger is better (Reward), degradation is defined as a relative drop:

$$\Delta_{d,\text{Reward}}(\%) = \frac{x_{d,\text{Reward}}^{\text{RL-RA}} - x_{d,\text{Reward}}}{x_{d,\text{Reward}}^{\text{RL-RA}}} \times 100.$$

For metrics where smaller is better (Regret, Violation, and CR), degradation is defined as a relative increase:

$$\Delta_{d,m}(\%) = \frac{x_{d,m} - x_{d,m}^{\text{RL-RA}}}{x_{d,m}^{\text{RL-RA}}} \times 100$$

where $m \in \{\text{Regret, Violation, CR}\}$. By construction, $\Delta_{d,m} > 0$ indicates worse performance than RL-RA, while $\Delta_{d,m} < 0$ indicates an improvement over RL-RA under the same metric.

B. Overall Comparison

Experimental results in Table I indicate that RL-RA achieves markedly improved feasibility and learning efficiency, with a constraint violation rate of approximately 1.1%–1.5% compared with 5.1%–13.0% for the baselines, and a regret of 25.8–28.7 versus at least 89.4 for all comparative methods. To ensure that this performance gap is not an artifact of unequal calibration, we emphasize that all baseline methods were subjected to explicit hyperparameter tuning that incorporated constraint satisfaction as a first-class objective; the tuning search space, validation criterion, and selected parameters are reported in the experimental setup. For methods that do not natively support constraint-aware training, we additionally report results under their best feasible operating points identified by a constraint-prioritized validation sweep, and we include a sensitivity analysis to demonstrate that the relative ranking is stable under reasonable variations of the constraint-related hyperparameters.

C. Comparison Under Uncertainty

To evaluate robustness under uncertain industrial observations, we inject Gaussian noise with standard deviation $\sigma \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$ into the input states of all datasets and compare five methods. Fig. 2 shows that RL-RA is consistently the most stable as σ increases: it achieves the highest cumulative reward with a slower degradation trend, particularly on 5GNet and IoTEnergy. The safety advantage is also clear, as RL-RA maintains the lowest violation rate across all noise levels, whereas several baselines exhibit sharp increases under moderate perturbations. In addition, RL-RA yields the smallest regret in all settings, with curves that rise gradually and remain below competing methods, indicating sustained decision quality under disturbance. These consistent trends across datasets

validate the robustness and constraint compliance of RL-RA in noisy industrial environments.

D. Ablation Analysis

Table II and Fig. 3 report an ablation study over three datasets using normalized cumulative reward, regret, violation rate (fraction of constraint-violating steps), competitive ratio, and degradation. Removing adversarial training consistently reduces reward (e.g., 938.1 to 849.2 on 5GNet) while increasing regret and violation rate, indicating that adversarial perturbations improve robustness under uncertainty. Eliminating feasibility projection yields the most severe safety degradation, with the violation rate increasing from 0.04 to 0.19 on 5GNet and from 0.03 to 0.22 on Alibaba, confirming that feasibility-aware action construction is essential for constraint compliance. Removing forecast embedding also degrades performance across datasets, suggesting that short-horizon predictive context improves decision quality under time-varying workloads. Overall, the full RL-RA configuration achieves the best trade-off, attaining the highest reward and lowest regret and violation rates across all benchmarks.

VI. CONCLUSION AND FUTURE WORK

RL-RA is a reinforcement learning based resource allocation framework for Industrial Cyber-Physical Systems that formulates allocation as a constrained stochastic control problem and combines predictive context, feasibility-aware action mapping, and adversarial disturbance handling to improve robustness under uncertainty and hard operational constraints. Theoretical results establish guarantees on regret, feasibility, and competitive performance, and experiments on three real-world datasets show consistent gains over baselines in cumulative reward, violation rate, and competitive ratio, with ablations confirming the importance of forecast embedding, adversarial training, and feasibility projection. Future extensions include multi-agent coordination, transfer learning across industrial domains, integrating model-based planning for improved sample efficiency and interpretability, and real-time deployment studies under practical compute and communication constraints.

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